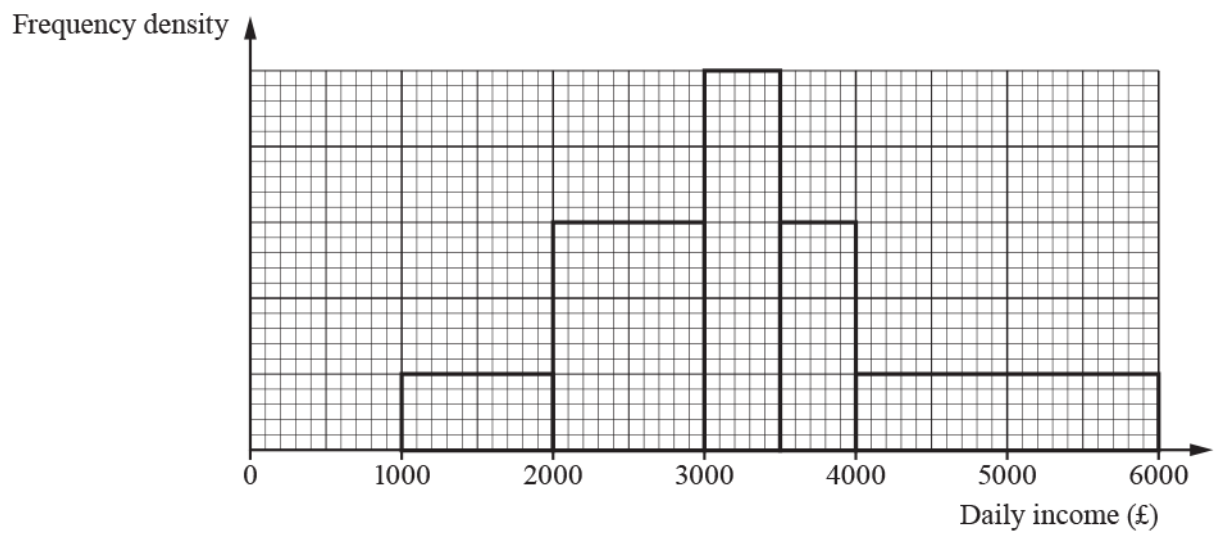


- 1 The masses, in grams, of apples of a certain kind have the distribution $N(67, \sigma^2)$. It is given that half of the apples have masses between 62 g and 72 g.

Determine σ .

[5]

- 2 The finance department of a retail firm recorded the daily income each day for 300 days. The results are summarised in the histogram.



(a) Find the number of days on which the daily income was between £4000 and £6000. [3]

(b) Calculate an estimate of the number of days on which the daily income was between £2700 and £3600. [3]

(c) Use the midpoints of the classes to show that an estimate of the mean daily income is £3275. [2]

An estimate of the standard deviation of the daily income is £1060. The finance department uses the distribution $N(3275, 1060^2)$ to model the daily income, in pounds.

- (d) Calculate the number of days on which, according to this model, the daily income would be between £4000 and £6000.

[2]

- (e) It is given that approximately 95% of values of the distribution $N(\mu, \sigma^2)$ lie within the range $\mu, \pm 2\sigma$. [2]
Without further calculation, use this fact to comment briefly on whether the proposed model is a good fit to the data illustrated in the histogram.

- 3 The masses, X grams, of tomatoes are normally distributed. Half of the tomatoes have masses greater than 56.0 g and 70% of the tomatoes have masses greater than 53.0 g.

(a) Find the percentage of tomatoes with masses greater than 59.0 g. [2]

(b) Find the percentage of tomatoes with masses greater than 65.0 g. [4]

(c) Given that $P(a < X < 50) = 0.1$, find a .

[3]

- 4 The heaviest 17% of rococo apples are classified as large, and the lightest 17% are classified as small. The remainder are classified as medium. The limits within which the masses of medium rococo apples lie are 96 g and 120 g. Stating a necessary assumption, estimate the mass of the heaviest rococo apple. [4]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1			$P(Y < 72) = 0.75$ $\Phi^{-1}(0.75)$ or 0.674 $\frac{72-67}{\sigma}$ $\frac{72-67}{\sigma} = 0.674$ $\sigma = 7.41$ or 7.42 (3 sf) Trial and Improvement $\Phi^{-1}(0.75)$ or 0.674 or $\Phi^{-1}(0.25)$ or -0.674 eg $\sigma = 8$: $67 - 8 \times 0.674 = 61.6$ $\sigma = 7$: $67 - 7 \times 0.674 = 62.3$ $\sigma = 7.41$: $67 - 7.41 \times 0.674 = 62.0$ $\Rightarrow \sigma = 7.41$ or $\sigma = 7.42$: $67 - 7.42 \times 0.674 = 62.0$ $\Rightarrow \sigma = 7.42$	M1(AO3.1b) M1(AO2.4) M1(AO2.1) A1(AO1.1) A1(AO1.1) M2 M1 A1 A1 [5]	oe May be implied, eg on diagram ± 0.674 implies M1M1 Allow 0.67 oe, eg 5 = 0.674σ A1 for correct equn, allow 0.67 SC correct answer with no working or irrelevant working: SC B3 (because "determine" rather than "find") May be implied At least one correct trial Trials leading to values either side of 62 Correct trial using $\sigma = 7.41$ or 7.42 and conclusion $\sigma = 7.41$ or 7.42 Examiner's Comments Many candidates recognised that they needed to use either $\Phi^{-1}(0.25)$ or $\Phi^{-1}(0.75)$ and went on to obtain the correct answer. Some others used $\Phi^{-1}(0.5)$, obtaining no marks. No candidate attempting a Trail and Improvement method showed sufficient detail to be awarded full marks and generally scored 3 out of 5. See published mark scheme for example of level of detail required.
			Total	5	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
2		a	Total area = 500 small squares $\frac{100}{500} \times 300$ = 60 (days)	B1 (AO1.2) M1 (AO1.1a) A1 (AO1.1) [3]	or 20 cm ² or other units or $\frac{4}{20} \times 300$ or equivalent	May be implied
		b	$\frac{3 \times 15 + 5 \times 25 + 1 \times 15}{500} \times 300$ = 111	M1 (AO2.1) M1 (AO1.1) A1 (AO1.1) [3]	M1 for denom & one term in num M1 M1 for correct $\times$ 300	oe in other units
		c	Frequencies: 30, 90, 75, 45, 60 $\frac{\Sigma fx}{300} = \frac{982500}{300}$ (= 3275 AG)	M1 (AO3.1a) A1 (AO1.1) [2]	Allow multiples of these correctly obtain 3275	
		d	$P(4000 < x < 6000) \times 300$ = $0.2419 \times 300 = 72.57$ so 73 days	M1 (AO3.4) A1 (AO1.1) [2]	Attempted, using $N(3275, 1060^2)$ BC accept truncation to 72 days	
		e	$3270 + 2 \times 1060 = 5395$ In histogram, well over 2.5% of values are above 5395, so model not a good fit	B1 (AO3.4) E1 (AO2.2b) [2]		
			Total	12		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3		a	$\mu = 56$ Percentage with masses > 59 g = 30%	B1(AO 1.1a) B1(AO 1.1) [2]	or 0.3
		b	$\Phi\left(\frac{53-56}{\sigma}\right) = 0.3, \frac{53-56}{\sigma} = -0.5244$ $\sigma = 5.721$ $X \sim N(56, '5.721^2)$ soi $P(X > 65) = 0.0578$ or 5.78% (3 sf)	M1(AO 2.1) A1(AO 1.1) M1(AO 2.4) A1(AO 1.1) [4]	or $P(X > 65) = P(z > \frac{65-56}{5.721})$ ft their σ $= P(z > 1.573)$ Or BC
		c	$P(X < 50) = 0.1471$ $P(X < a) = 0.0471$ $a = 46.4$ (3 sf)	M1(AO 1.1a) A1(AO 2.1) A1(AO 1.1) [3]	
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
4			Assume masses normally distr. 66% of masses lie approximately within $\mu \pm \sigma$ and greatest mass $\approx \mu \pm 3\sigma$ sd = 0.5(120 – 96) (= 12) Greatest mass = 120 + 2×12 = 144 (g)	B1(AO 1.2) M1(AO 3.1b) M1(AO 1.1) A1(AO 1.1) [4]	or similar both stated or implied or greatest mass = $\frac{96+120}{2} + 3 \times 12 = 144$	Allow 144 to 145
			Total	4		